EcoHarvest - Intelligent Bio-Waste Processing System with Thermal Optimization and IoT Monitoring



Figure 1: Schematic illustration showing the overall workflow of the *EcoHarvest* system

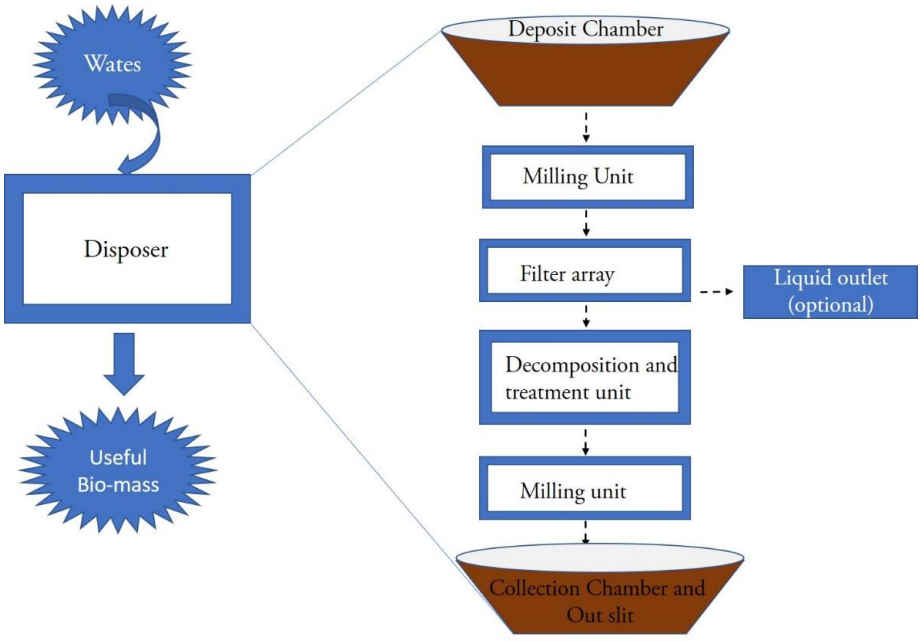


Figure 2: Functional architecture of the proposed *EcoHarvest* system illustrating the sequential processing stages

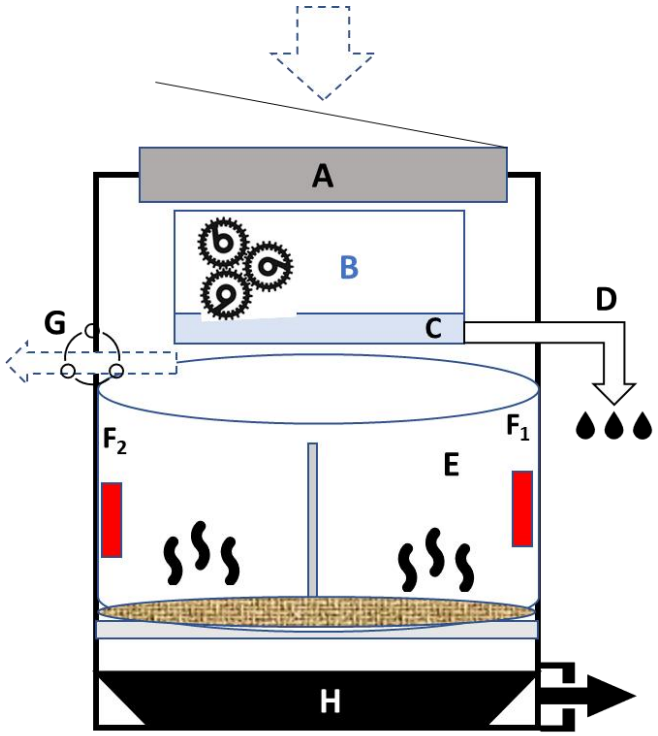


Figure 3: Schematic diagram of the proposed bio-waste processing unit showing the major functional compartments.

A: Lid (Waste disposal area)

B: Grinding area, here the bulk solid waste will crush into small pieces.

C: Elimination of liquid, here the liquid waste will be removed, and the solid filtrate will collect at the bottom

D: Liquid outlet

E: Drying area, this area will remove moisture from the filtrates and make it dry

F1 & F2: IR Lamb, this will dehydrate the remaining filtrate and dry

G: Exhaust, to eliminate the vapour,

H: The bottom tray is a collecting unit of biomass

A. Grinding Unit

The grinding unit is the first stage of the bio-waste disposer. Once the waste is deposited, it is processed by high-grade stainless steel blades that reduce large organic materials into smaller, uniform particles. This ensures that the waste can be efficiently dried and processed in the following stages. The grinding process also helps separate liquid and solid components, with the liquid being drained away at this stage to optimize drying later on.

B. Drying Unit

The drying unit is where most of the moisture is removed from the waste. Hot air is circulated around the waste using an internal heating coil, and the waste is stirred continuously to promote even drying. A motor-driven central paddle rotates the waste, ensuring thorough air exposure. The drying unit significantly reduces the overall mass and volume of the waste before it moves to the next stage.

C. IR Lamp

The IR (Infrared) lamp is an essential component that further accelerates the drying process by removing any residual moisture after the hot air treatment. The IR lamp is adjustable, allowing the intensity to be modified depending on the type and quantity of waste. This ensures that the final biomass is dry and odorless, ready to be collected and used as fertilizer.

D. Exhaust System

An exhaust system is fitted at the top of the device to manage airflow and odor control. Hot, moist air is vented out, preventing the build-up of unpleasant smells. The exhaust includes a filtration system to trap any odors or particulate matter, ensuring the surroundings remain clean and odor-free. This system ensures a hygienic and pleasant user experience.

E. Biomass Collection Tray

Once the waste has been dried and treated, the final product—dry, odorless biomass—collects in a tray at the bottom of the device. This biomass can be directly used as fertilizer for home gardens or small-scale agricultural purposes. The tray is easy to remove, clean, and replace for continuous operation.